

GOOD SOIL STEM ACADEMY — MASTER KIT SPECIFICATION & PROCUREMENT TABLE

Complete 25-Class Kit Inventory for Junior Batch (Ages 6–9 / Grades 1–4) & Senior Batch (Ages 10–14 / Grades 5–9)

Program Metric	Junior Batch (Ages 6–9 / Grades 1–4)	Senior Batch (Ages 10–14 / Grades 5–9)	Procurement & Logistics Architecture
Curriculum Scope	25 Hands-on Classes (50 Hours instruction)	25 Hands-on Classes (50 Hours instruction)	Weekly 2-hour weekend modules across 6 months
Kit Distribution	10 Take-Home Kits + 10 Group Sets + 4 Challenges + 1 Expo	10 Take-Home Kits + 10 Group Sets + 4 Challenges + 1 Expo	Take-home kits retained by students; Studio kits reused
Domain Coverage	Aerodynamics, Basic Circuits, Gears, Bionics, Optics, Science	Hydraulics, Advanced Aero, Electronics, Sensors, Autonomous Bots	7 Verified Direct Vendors (Pludo, Mindtronix, Kiddale, etc.)

1. Junior Batch (Ages 6–9 / Grades 1–4) — Complete 25-Class Kit List

Focus: Tactile physical mechanics, basic circuits, introductory aerodynamics, optics, chemical reactions, and bionic linkages. (10 Take-Home Giveaway Kits + 10 Group Studio Kits + 4 Milestones + 1 Expo)

Cls	Kit / Activity Name	Domain	Format / Type	Vendor	SKU / Code	Unit Price	Core Learning Concept & Key Pedagogical Outcome
01	Electric Propeller & Wind Car	Electricity I	Take-Home #1	Pludo (Robocraze)	WEB-PLU-015	Rs. 467	Basic DC motor circuits, propeller thrust, aerodynamics, switch control.
02	High-Lift Paper Gliders & Launcher	Aero I	Group Kit #1	Good Soil Aero	CUSTOM-AERO-01	Rs. 150	Airfoil cross-sections, lift vs gravity, thrust launcher, wing aspect ratio.
03	Kitchen Chemistry & Color Fireworks	Science I	Group Kit #2	Butterfly Edufields	WEB-BUT-051	Rs. 649	Acid-base neutralization, CO2 gas generation, liquid density layers, color kinetics.
04	Wooden Catapult & Trajectory Shooter	Mechanics I	Take-Home #2	Practiko	WEB-PRA-052	Rs. 499	Class 3 levers, elastic potential energy, fulcrum position, trajectory angles.
05	Spin Art & Craft STEAM Machine	STEAM Art I	Take-Home #3	Butterfly Edufields	WEB-BUT-060	Rs. 399	Centrifugal force, rotational dynamics, symmetry, electric motor speed blending.
06	Assessment 1: Island Escape Challenge	Challenge 1	Assessment	Good Soil Studio	STUDIO-ASSESS-01	Rs. 200	Flotation engineering challenge & catapult target accuracy quest (Badge: Explorer).
07	Wooden Paddle Steamer Propeller Boat	Mechanics II	Take-Home #4	Kiddale	WEB-KDA-031	Rs. 424	Water displacement, paddle propulsion, buoyant force, rotary energy conversion.
08	Balsa/Foam Chuck Glider & Flight Tuning	Aero II	Group Kit #3	Good Soil Aero	CUSTOM-AERO-02	Rs. 200	Center of Gravity (CG) balancing, elevator/rudder trim tabs, dihedral stability.
09	Wooden Glow Table Lamp & Switch Circuit	Electricity II	Take-Home #5	Kiddale	WEB-KDA-042	Rs. 424	Closed vs open loops, LED anode/cathode polarity, conductor continuity, woodcraft assembly.
10	Wooden Mechanical Carousel & Gear Train	Mechanics III	Group Kit #4	Kiddale	WEB-KDA-032	Rs. 424	Compound spur gear trains, speed reduction vs torque increase, bevel gear direction.
11	Morse Code Telegraph & Buzzer Station	Electronics I	Take-Home #6	Studio / DIY	STUDIO-ELEC-01	Rs. 150	Electromagnetic buzzers, momentary push switches, Morse code encoding & decoding.
12	Assessment 2: STEAM Carnival Ride	Challenge 2	Assessment	Good Soil Studio	STUDIO-ASSESS-02	Rs. 200	Multi-gear motorized carnival ride build & team showcase (Badge: Master Crafter).
13	Pneumatic Air-Powered Rocket Launcher	Aero III	Take-Home #7	Pludo (Robocraze)	WEB-PLU-014	Rs. 370	Pneumatic pressure accumulation, sudden release valve, rocket fin stabilization.

ClS	Kit / Activity Name	Domain	Format / Type	Vendor	SKU / Code	Unit Price	Core Learning Concept & Key Pedagogical Outcome
14	10-in-1 Ultimate Electricity Lab	Electronics II	Group Kit #5	Butterfly Edufields	WEB-BUT-030	Rs. 589	Series vs parallel connections, magnetic reed switches, conductor vs insulator testing.
15	Mechanical Secret Cipher Wheel & Maze	Coding I	Take-Home #8	Practiko	WEB-PRA-068	Rs. 499	Caesar shift cipher, substitution algorithms, binary secret encoding, logical decision trees.
16	3-Ball Motorized Sun, Earth & Moon	Space Science	Group Kit #6	Kiddale	WEB-KDA-139	Rs. 449	Orbital mechanics, solar/lunar eclipses, diurnal rotation, seasonal axial tilt.
17	Pin Hole Camera & Optical Periscope	Science II	Take-Home #9	Practiko	WEB-PRA-054	Rs. 499	Rectilinear propagation of light, focal inverted projection, 45-degree mirror reflection.
18	Assessment 3: Sky & Signals Derby	Challenge 3	Assessment	Good Soil Studio	STUDIO-ASSESS-03	Rs. 200	Precision rocket range trials + optical mirror signal decoding race (Badge: Flight Cadet).
19	Wooden Walking T-Rex Bionic Robot	Robotics I	Group Kit #7	Kiddale	WEB-KDA-039	Rs. 594	Four-bar linkage kinematics, crank-rocker motion, biomimetic 2-legged walking balance.
20	Electric Coin-Eating Smart Bank Robot	Robotics II	Group Kit #8	Kiddale	WEB-KDA-133	Rs. 599	Automated mechanical switch triggers, dual-jaw linkage, motor stall torque.
21	Wooden Solar Power Rover Car	Clean Energy	Take-Home #10	Kiddale	WEB-KDA-043	Rs. 594	Photovoltaic energy conversion, solar panel angle tuning, direct motor drive efficiency.
22	Two-Wheeled Obstacle Rover Team Build	Robotics III	Group Kit #9	Kiddale	WEB-KDA-140	Rs. 499	Differential drive steering, dual-motor synchronization, chassis center of gravity.
23	Mini Quad Drone Discovery & Pilot	Drones	Group Kit #10	BrainHap / Studio	WEB-BRH-003	Rs. 2,999	Quad-rotor aerodynamics, roll/pitch/yaw control, gyro stabilization, pilot flight arena.
24	Assessment 4: Future City Prototype	Challenge 4	Assessment	Good Soil Studio	STUDIO-ASSESS-04	Rs. 250	Capstone integration: Combine bionic mechanisms + solar + circuits (Badge: Young Innovator).
25	Grand Expo: Medals & Certs	Graduation	Grand Expo	Good Soil Studio	STUDIO-EXPO-01	Rs. 300	Live flight arena, robot racing, parent demo day, STEM Passport stamps & medals.

2. Senior Batch (Ages 10–14 / Grades 5–9) — Complete 25-Class Kit List

Focus: Advanced fluid hydraulics, projectile physics, sensor integration (acoustic/strain/ultrasonic), motor polarity logic, bionic multi-leg locomotion, and drone avionics. *(10 Take-Home Giveaway Kits + 10 Group Studio Kits + 4 Milestones + 1 Expo)*

Cls	Kit / Activity Name	Domain	Format / Type	Vendor	SKU / Code	Unit Price	Core Learning Concept & Key Pedagogical Outcome
01	Hydraulic Mech Gripper Robot Arm	Mechanics I	Take-Home #1	Kintaro DIY	WEB-KIN-001	Rs. 749	Pascal's principle, fluid mechanical advantage, master/slave hydraulic cylinders, 2-DOF gripper.
02	Rubber-Band High-Endurance Monoplane	Aero I	Group Kit #1	Good Soil Aero	CUSTOM-AERO-03	Rs. 400	Cambered airfoil aerodynamics, rubber motor torque curve, propeller pitch optimization.
03	150+ Chemistry & Physics Lab	Science I	Group Kit #2	Butterfly Edufields	WEB-BUT-052	Rs. 669	Exothermic & endothermic reactions, universal pH titration, precipitation, crystal nucleation.
04	Precision Force Vector Catapult	Mechanics II	Take-Home #2	Mindtronix	WEB-MIN-003	Rs. 712	Parabolic kinematics, projectile velocity vectors, adjustable tension stop angles, target math.
05	SolarGlide Kinetic Solar Aircraft	STEAM Art I	Take-Home #3	Pludo (Robocraze)	WEB-PLU-024	Rs. 719	Rotary solar counterweight balance, low-friction bearing pivot, kinetic shadow aerodynamics.
06	Assessment 1: Medieval Siege Challenge	Challenge 1	Assessment	Good Soil Studio	STUDIO-ASSESS-01S	Rs. 250	Vector force target competition + chemical indicator identification sprint (Badge: Apprentice Engineer).
07	DIY Electronic Weighing Scale	Electronics I	Take-Home #4	Mindtronix	WEB-MIN-017	Rs. 623	Piezoresistive strain gauge deflection, Wheatstone bridge concept, tare calibration logic.
08	Large Wingspan Foam Glider & CG Trim	Aero II	Group Kit #3	Good Soil Aero	CUSTOM-AERO-04	Rs. 500	Glide ratio (L/D) maximization, aerodynamic stall mitigation, lateral dihedral stability.
09	2-Way Reversible Motor Electric Car	Electronics II	Take-Home #5	Pludo (Robocraze)	WEB-PLU-064	Rs. 710	DPDT rocker switch polarity inversion, H-bridge fundamentals, drivetrain traction tuning.
10	Heavy-Duty Cable Car Overhead Lift	Mechanics III	Group Kit #4	Mindtronix	WEB-MIN-015	Rs. 759	Compound pulley systems, mechanical tension vectors, high-torque reduction gearbox.
11	Neon Glow Optical Infinity Mirror	STEAM Art II	Take-Home #6	Studio / DIY	STUDIO-ART-01S	Rs. 350	Two-way semi-reflective mirrors, recursive light ray geometry, LED perimeter array.
12	Assessment 2: Rescue Rover Canyon Race	Challenge 2	Assessment	Good Soil Studio	STUDIO-ASSESS-02S	Rs. 250	Cable-driven payload transfer across obstacle gorge & reverse polarity time trial (Badge: Systems Builder).
13	Motorized Electric DIY Aircraft	Aero III	Take-Home #7	Pludo (Robocraze)	WEB-PLU-074	Rs. 629	High-RPM coreless motor thrust, lithium capacitor discharge rate, powered climb profiles.
14	Motorized Forklift & Heavy Cargo Lifter	Mechanics IV	Group Kit #5	Mindtronix	WEB-MIN-007	Rs. 702	Self-locking worm drive gearboxes, vertical rack-and-pinion carriage, load center stability.
15	Clap-Racer Sound-Activated Car	Sensors I	Take-Home #8	Pludo (Robocraze)	WEB-PLU-009	Rs. 719	Electret microphone transducer, transistor trigger latch, acoustic noise threshold trimming.
16	Orbital Satellite & Telemetry Antenna	Space Tech	Group Kit #6	Pludo (Robocraze)	WEB-PLU-019	Rs. 629	Geostationary vs LEO satellite orbits, solar panel orientation tracking, signal coverage cones.
17	Infinity Curve & Harmonic Pendulum	Science II	Take-Home #9	Kintaro DIY	WEB-KIN-014	Rs. 799	Coupled harmonic oscillators, phase relationship equations, resonance damping, Lissajous curves.
18	Assessment 3: Flight Derby & Sensor Maze	Challenge 3	Assessment	Good Soil Studio	STUDIO-ASSESS-03S	Rs. 250	Powered aircraft duration trial + sound-triggered obstacle navigation course (Badge: Avionics Specialist).
19	Multi-Axis Motorized Robot Claw Machine	Robotics I	Group Kit #7	Pludo (Robocraze)	WEB-PLU-075	Rs. 629	3-axis Cartesian mechanical linkages, bidirectional motor control, end-effector slip friction.

Cls	Kit / Activity Name	Domain	Format / Type	Vendor	SKU / Code	Unit Price	Core Learning Concept & Key Pedagogical Outcome
20	Bujji Autonomous Obstacle Robot	Robotics II	Group Kit #8	Mindtronix	WEB-MIN-022	Rs. 799	Ultrasonic time-of-flight echo sensing, obstacle avoidance decision state machine.
21	Programmable Motorized Arrow Launcher	Logic & Coding	Take-Home #10	Mindtronix	WEB-MIN-006	Rs. 786	Cam-driven rapid fire release, microswitch sequence timing, mechanical automation safety.
22	Moto-Bot Bionic Multi-Leg Walker	Robotics III	Group Kit #9	Kintaro DIY	WEB-KIN-015	Rs. 799	Theo Jansen & Klann linkage gait synthesis, multi-leg coordinate phase shifting, terrain traversal.
23	BrainHap HD Camera Drone Piloting	Drones	Group Kit #10	BrainHap	WEB-BRH-003	Rs. 2,999	6-axis gyroscope flight trim, optical flow altitude hold, FPV telemetry live stream diagnostics.
24	Assessment 4: Shark Tank Defense	Challenge 4	Assessment	Good Soil Studio	STUDIO-ASSESS-04S	Rs. 300	Engineering prototype defense, circuit logic explanation, live performance demo (Badge: Master Innovator).
25	Senior Tech Symposium & Expo	Graduation	Grand Expo	Good Soil Studio	STUDIO-EXPO-02	Rs. 350	Public technical symposium, autonomous robot battle arena, flight exhibition, trophies.

3. Optional Supplementary Kits for Early Learners (Ages 3–6 / Preschool)

Modular early-childhood activity sets available for introductory weekend workshops or sibling daycare engagement.

#	Kit / Resource Name	Age Group	Domain	Vendor	SKU Code	Unit Price	Kit Description & Educational Focus
1	155 Alphabets & Words Magnet Lab	3–6 yrs	Literacy & Phonics	Butterfly Edufields	WEB-BUT-063	Rs. 549	Soft EVA foam magnetic letter blocks, spelling flashcards, phonics word builder.
2	90+ pcs Magnetic Numbers & Shapes	3–5 yrs	Early Math & Logic	Butterfly Edufields	WEB-BUT-061	Rs. 499	Geometric shape recognition, tactile number puzzles, early addition/subtraction.
3	Air Magic Bernoulli's Law Early Kit	4–6 yrs	Early Physics	Pludo (Robocraze)	WEB-PLU-032	Rs. 475	Soft continuous airflow levitation ball, safe toddler-proof fan nozzle design.
4	Little Builders Cable Bridge Project	4–6 yrs	Early Engineering	Pludo (Robocraze)	WEB-PLU-030	Rs. 475	Chunky wooden structural beams, large interlocking joints, bridge load testing.
5	4-in-1 Animals & Dinosaurs Jigsaw	3–6 yrs	Visual Cognition	Butterfly Edufields	WEB-BUT-010	Rs. 399	Thick layered wooden animal anatomy puzzles, fine motor skill coordination.

4. Primary Vendor Procurement Directory

Direct verified manufacturers and wholesale distributors for bulk request-for-quotation (RFQ).

Vendor / Supplier	Total Kits	Priority	Key Kit Lines in Curriculum	Official Store / Portal
Pludo (Robocraze)	7 Kits	Tier 1	Motorized Wind Car, Pneumatic Rocket, SolarGlide, Clap-Racer, Motorized Plane, Robot Claw	robocraze.com
Mindtronix	7 Kits	Tier 1	Precision Catapult, Electronic Scale, Cable Car, Forklift, Bujji Robot, Arrow Launcher	mindtronix.in
Kiddale	8 Kits	Tier 1	Paddle Steamer, Glow Lamp, Carousel, Sun-Earth-Moon, T-Rex Walker, Coin Bank, Solar Rover	kiddale.com
Butterfly Edufields	4 Kits	Tier 1	Spin Art Machine, 10-in-1 Electricity Lab, Kitchen Chemistry, 150+ Physics/Chem Lab	butterflyfields.com
Practiko	3 Kits	Tier 1	Wooden Catapult, Secret Cipher Wheel & Maze, Pin Hole Camera & Optical Periscope	practiko.in
Kintaro DIY	3 Kits	Tier 2	Hydraulic Mech Gripper Arm, Harmonic Motion Pendulum, Moto-Bot Bionic Multi-Leg Walker	kintarodiv.com
BrainHap	1 Kit	Tier 1	Beginner HD Camera Quadcopter Drone Kit (Demonstration & Pilot Station)	brainhap.com